



Document Management System



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Title: Inotropic Support for Critically Ill Children

Introduction:

In children with hypotension (SBP < 10th centile) , start with volume loading 20 ml/kg of crystalloids , or 10ml/kg ofcolloids

Volumes up to 60ml/kg crystalloids during 1st hour ofresuscitation can be used.

Note that in children withcongestive heart failure or volume overload , careful volume administration (5ml/kg) adviced with assessment for increase in liver size or lung crepitations

- You can start with peripheral dopamine , consider starting dobutamine forchildren with underlying heart disease .

(consider adding 2ndinotrope if dopamine or dobutamine > 10 mcg/kg/min and the child will need acentral access)

- Involve the consutant on duty early in the management and disscuss theinotropic plan .

Guide on inotropic support choice

*(derivedfrom surviving sepsis campaign international 2012)

Cold shock/Normal blood pressure

- Add vasodilator with volume loading (nitrosvasodilators , milrinone)

Cold shock / low bloodpressure

- Titrate fluid and adrenaline infusion

- Consider noreadrenaline if still hypotensive

Warm shock / low bloodpressure

If still hypotensive , consider vasopressin

Commonly used inotropic agents

Dopamine :

Naturally occurring drug , has alpha, beta and dopa effect , act on dopa 1 receptors (mediate vasodilatation) , dopa 2 receptors (mediate vasoconstriction) , precursor of noreadrenaline

Dose 2.5-10 mcg/kg/min B1 effect , inotropic , chronotropic actions

Dose >10mcg/kg/min alpha , beta action (inotrope ,chronotrope , vasopressor)

(note that doses >10mcg/kg/min induce peripheral vasoconstriction)

Dobutamine

Beta agonist effect , + inotropic effect , can lead to tachycardia (higher doses lead to chronotropic effect) Systemic and pulmonary vasodilator from beta 2 effect

Adrenaline

Naturally occurring alpha , beta 1 and beta 2 effect

Doses 0.05-0.1 mcg/kg/min : beta 1 and beta 2 effect predominate

0.1 mcg/kg/min lead to alpha effect in addition with peripheral vasoconstriction

Noreadrenaline

Naturally occurring alpha and beta 1 agonist , precursor of adrenaline

Beta 1 effect lead to positive inotropic effect while alpha effect lead to vasopressor effect , used in conditions of warm shock

Milrinone

Action : type 111 phosphodiesterase inhibitor

Non glycoside , non catecholamine inotrope with additional vasodilatory and lusitropic effect , act via increasing CAMP levels and therefore increase Ca^{++} transport across the cell , it is used in children in post open heart surgery and children with low cardiac output state following cardiac conditions.

Levosimendan :

Calcium sensitizer , binds to cardiac troponin c leading to actin-myosin cross -bridges increasing cardiac contractility and performance without increasing energy demands by the myocardium

Agent	Doses (IV)	Peripheral Vascular Effect	Cardiac Effect	Conduction System Effect
NONCATECHOLAMINES				
Digoxin (total digitalizing dose)	20 µg/kg premature 30 µg/kg neonate (0–1 mo) 40 µg/kg infant (< 2 yr) 30 µg/kg child (2–5 yr) 20 µg/kg child (> 5 yr)	Increase peripheral vascular resistance 1–2+; acts directly on vascular smooth muscle	Inotropic effect 3–4+; acts directly on myocardium	Slows sinus node slightly; decreases AV conduction more
Calcium chloride	10–20 mg/kg/dose (slowly)	Variable; age-dependent; vasoconstrictor	Inotropic effect 3+; depends on ionised Ca ²⁺	Slows sinus node; decreases AV conduction
Gluconate	50–100 mg/kg/dose (slowly)			Reflex tachycardia
Nitroprusside	0.5–5 µg/kg/min	Donates nitric oxide group to relax smooth muscle and dilate pulmonary and systemic vessels	Indirectly increases cardiac output by decreasing afterload	
Nitroglycerin	0.5–10 µg/kg/min	Primarily venodilator, as a nitric oxide donor may cause pulmonary vasodilation and enhance coronary vasoreactivity after aortic cross-clamping	Decreases preload; may decrease afterload; reduces myocardial work related to change in wall stress	Minimal
Milrinone	50–75 µg/kg loading dose 0.25–1.0 µg/kg/min	Systemic and pulmonary vasodilator Potent vasoconstrictor	Diastolic relaxation (lusitropy); measurable inotropic effect	Minimal tachycardia

Summary of Selected Vasoactive Agents (Continued)

Agent	Dose Range	α	β ₂	ε	β ₁	β ₂	Comment
Isoproterenol	0.05–0.5 µg/kg/min	0	4+	0	4+	4+	Strong inotropic and chronotropic agent; peripheral vasodilator; reduces preload; pulmonary vasodilator; limited by tachycardia and oxygen consumption
Norepinephrine	0.1–0.5 µg/kg/min	4+	0	0	2+	0	Increases systemic resistance; moderately inotropic; may cause renal ischemia
Epinephrine	0.03–0.1 µg/kg/min 0.2–0.5 µg/kg/min	2+ 4+	1–2+ 0	0 0	2–3+ 4+	2+ 3+	Beta ₂ effect with lower doses; best for blood pressure in anaphylaxis and drug toxicity
Dopamine	2–4 µg/kg/min 4–8 µg/kg/min >10 µg/kg/min	0 0 2–4+	0 2+ 0	2+ 2+ 0	0 1–2+ 1–2+	0 1+ 2+	Splanchnic and renal vasodilator; may be used with isoproterenol; increasing doses produce increasing α effect
Dobutamine	2–10 µg/kg/min	1+	2+	0	3–4+	1–2+	Less chronotropy and arrhythmias at lower doses; effects vary with dose similar to dopamine; chronotropic advantage compared with dopamine may not be apparent in neonates
Fenoldopam	0.05–1 µg/kg/min (see text; little experience in children)						Powerful D ₁ agonist; little chronotropic or inotropic effect but may redistribute flow to renal bed and improve urine output

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Pediatric Critical Care Medicine (4 th edi)	David L.WesselPeter C Lausen	2011	401-440

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